

WEEKLY PROGRESS REPORT

Name:	Dharshini K	Domain:	Data Science and Machine Learning
Project Name:	Agriculture Production Prediction	Week:	Final Week
Date of Submission:	10/07/2026		

I. Overview

In this final week, the primary focus was on completing the deployment of the traffic forecasting system and project wrap-up. The end-to-end forecasting pipeline was finalized and prepared for its live production environment. Rigorous stress testing and load testing were conducted on the deployed system. Final user interface enhancements to the dashboard were completed, and a comprehensive user guide technical documentation were compiled to facilitate a seamless transition and future maintenance.

II. Achievements

- Successfully deployed the XGBoost model as a web application on the target cloud platform (AWS/GCP/Azure).
- Implemented the real-time alert system to notify users of abnormal traffic conditions based on model forecasts.
- Performed comprehensive load testing to ensure the deployed system can handle peak-time traffic requests.
- Completed final dashboard UI/UX enhancements, adding data update frequency controls and detailed tooltip information.
- Generated all project documentation, including code comments, model performance logs, API specifications, and a user manual.
- Completed formal system handover to the operational team.
- Gathered and analyzed initial user feedback for final refinements.

III. Challenges

- Managing unexpected data pipeline lag during high-frequency data updates.
- Fine-tuning the alert thresholds to minimize false positives without missing critical events.
- Optimizing the cloud deployment for cost-effectiveness while maintaining performance.

IV. Learning Resources

- Cloud platform documentation (e.g., AWS SageMaker, Kubernetes).
- Stress testing tools documentation (e.g., Locust, Apache JMeter).
- Best practices for productionizing Machine Learning models.

V. Final Report and Project Closure

- Conducted the final presentation to project stakeholders.
- Archiving all code, data models, and artifacts.
- Documenting lessons learned and potential future improvements.
- Handover and final project sign-off.

VI. Additional Comments

This final week marks the successful conclusion of the Data Science and Machine Learning Traffic Forecasting project. The model is now deploy-ready and the system is fully operational. This experience has deepened my knowledge of the full machine learning lifecycle, from data acquisition to cloud-based deployment and operational monitoring. I look forward to applying these skills in future projects and to seeing the impact of this system in a real-world scenario.

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